

Qatar-1b

R-2396

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_111102 · created 2026-08-21T00:05:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455867.73741 +/- 0.00032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.011
Transit depth (Rp/Rs)^2	2.37 +/- 0.32 [%]
Orbital Inclination (inc)	84.24 +/- 0.89
Airmass coefficient 1 (a1)	1.257 +/- 0.014
Airmass coefficient 2 (a2)	-0.1134 +/- 0.004
Scatter in the residuals of the lightcurve fit is	1.14 %
Best Comparison Star	None
Optimal Aperture	2.9
Optimal Annulus	7.83
Transit Duration (day)	0.0702 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.154 $\pm$ 0.011 vs 0.1463 (deviation 0.54 σ)
Duration fit vs published	0.0702 d vs 0.0692 d (deviation 0.16 σ)
Mid-time O–C	0.9 min
Depth SNR	10.9
Residual scatter	1.14 %

Light curve

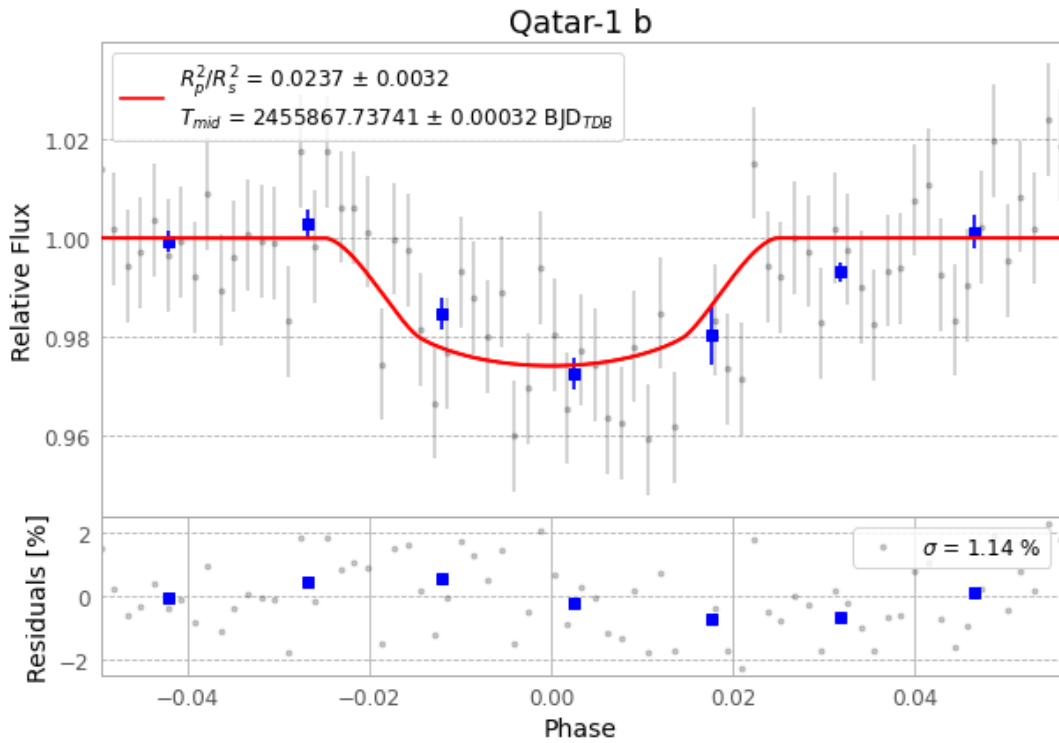


Figure 1 — FinalLightCurve_Qatar-1 b_2011-11-01.png (SHA-256 f72983a2ae243f5a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 246], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4769baa26cad3523...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), Qatar-1111102035712.FITS → Qatar-1111102073312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-111102.log (1218ff9604c1...), FinalLightCurve_Qatar-1 b_2011-11-01.png (f72983a2ae24...), FinalLightCurve_Qatar-1 b_2011-11-01.csv (60e0f0e7981a...)

Compute resources

Wall time 205.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:05Z.